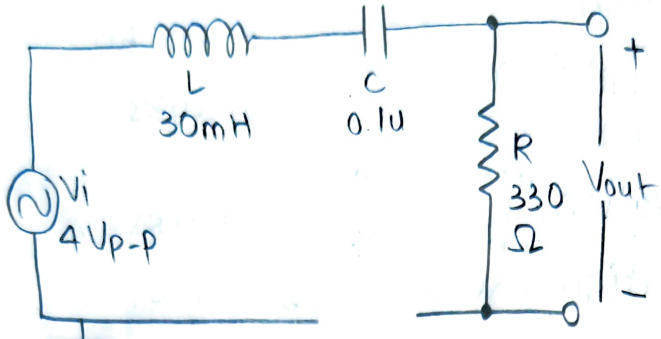
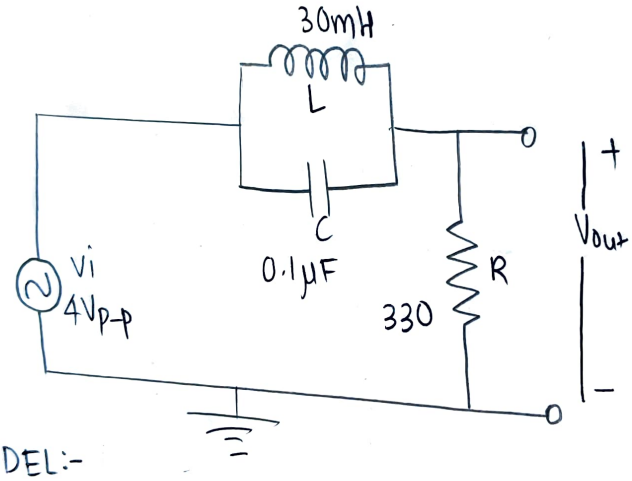
CIRCUIT DIAGRAM:-

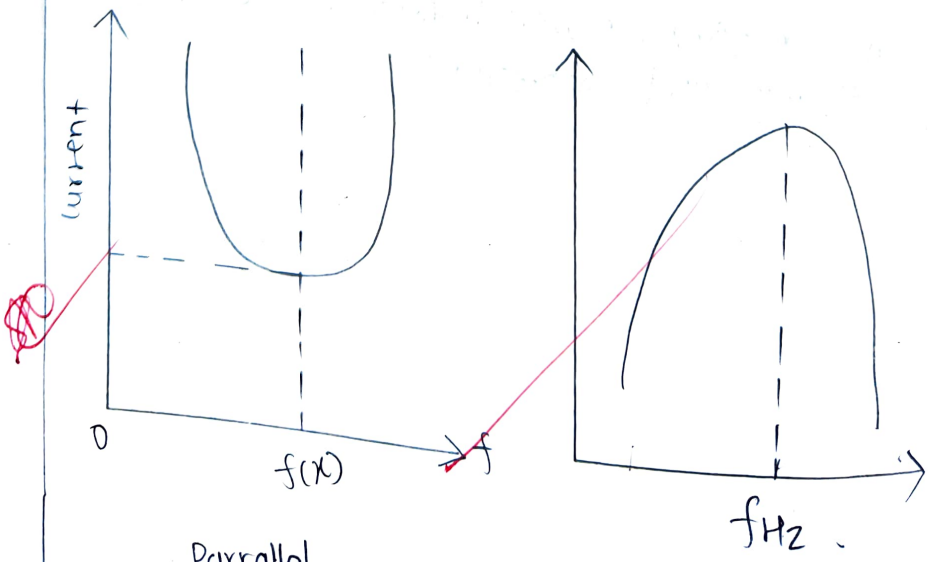


Series RLC circuit

Parallel RLC circuit



MODEL:-



Parrallel Resonance

Exp 7
Date:- 8/3/26

FREQUENCY REPONSE OF SERIES AND PARALLEL RESONANCE CIRCUITS

AIM:-

To study the behaviour of series and parallel RLC circuits at resonance and to determine the resonant frequency, bandwidth and Q of the given RLC circuit using Multisim simulator.

APPARATUS REQUIRED:-

S.no	Components	Specification	Quantity
1.	CRO	10MHz	1
2.	DIB	—	1
3.	Capacitors	0.1μf	1
4.	Resistors	330Ω	1
5.	Bread boards	—	—
6.	Connecting wires	—	—
7.	PC with Multism simulator	—	—

FORMULA:-

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$BW = f_2 - f_1$$

$$Q = \frac{f_0}{BW}$$

Tabulation = Vin = 4Vp-p.

Series Resonance			Parallel Resonance		
Frequency (Hz)	output voltage (V)	output current (mA)	Frequency (Hz)	output Voltage (V)	output current (mA)
500	0.8	2.4	500	3.9	11.8
1000	1.4	4.2	1000	3.8	11.5
1500	2.3	7.0	1500	3.6	10.9
2000	3.2	9.7	2000	3.3	10.0
2500	3.8	11.5	2500	2.8	8.5
2900	4.0	12.1	2900	2.2	6.6

Calculation:-

Resonant frequency = $f_0 = \frac{1}{2\pi\sqrt{LC}}$.

$$f_0 = \frac{1}{2\pi\sqrt{(30\times10^{-3})(0.1\times10^{-6})}}$$

$$= \frac{1}{2\pi\sqrt{3\times10^{-9}}}$$

$$= \frac{1}{2\pi(5.47\times10^{-5})} \Rightarrow 2900\,Hz.$$

Resonant frequency $(f_0) = 2.9\,kHz$.

O/P verified

Bandwidth Calculation:-

$$BW = \frac{R}{2\pi L} = \frac{330}{2\pi \times 0.03}.$$

$$= \frac{330}{0.188} \Rightarrow 1750.$$

Bandwidth = 1.75 kHz.

Q factor :-

$$Q = \frac{f_0}{BW}.$$

$$= \frac{2900}{1750} \Rightarrow 1.65.$$

Q factor = 1.65

MARKS SPLIT-UP :-

S.no	Description	Marks alloted	Marks obtained
1.	Experimental setup/ connections	20	20
2.	Observation/readings.	20	20
3.	Calculations/Results.	20	20
4.	Viva	20	20
5.	Record	20	20
		Total = 100	100

RESULT :-

10

Thus, the phenomenon of resonance in RLC circuit was studied and the following were determined using Multism simulator.

18/12

- Series RLC circuit :-
  Resonant frequency = 2.9 kHz
  Bandwodth = 1.75 kHz
  Q factor = 1.66

- Parallel RLC circuit :-
  Resonant frequency = 2.9 kHz
  Bandwodth = 1.75 kHz
  Q factor = 1.65